ECE473 Milestone II Update: New Rain Insurance Payouts and the AgriHedge App

Maya, Evan, and Jack • November 12, 2026

Summary. Milestone I showed that today's Kalshi rain contracts copy USDA's rainfall insurance (PRF) poorly and at a higher cost. In Milestone II we found out why, used that to design new insurance payouts from Kalshi's listed rain contracts, and built an app that quotes them for a farm. Below, *fit* is the share of a farm's year-to-year insurance payout swings that a hedge reproduces, measured on years it was not trained on.

1. Planned vs. actual progress

Jack *(planned: find the cheapest Kalshi portfolio that matches a farm's PRF payout; live pricing for the app)*

- **W4:** built an optimizer that picks contracts while accounting for fees, cash tied up until settlement, and how many contracts are actually for sale. Even the best portfolio reaches a fit of only about one third, so a better optimizer cannot fix the problem on its own.
- **W5:** built the backend that turns a farmer's plan into an order. It uses live Kalshi prices where contracts are listed and labels estimates for months without a live book.

Maya *(planned: explain where the remaining hedge error comes from; farmer screen)*

- **W4:** added real-world limits one at a time to measure each. Geography costs the most: Kalshi settles on an airport gauge 30 to 55 miles from the farm. Monthly contracts (PRF insures two-month windows) and thin order books come next. This separates what better contracts could fix from what they cannot.
- **W5:** built the farmer screen: click your land, see the nearest station, how well it tracks your farm, and the cost. Covering months outside a farmer's PRF selection reached median holdout fit 0.61, versus 0.23 for copying selected intervals; the app recommends that companion at 9 of 11 farms.

Evan *(planned: find the risks today's contracts cannot cover and propose fixes; test the app against PRF)*

- **W4:** distance to the airport gauge decides whether a hedge works. Across 150 random locations, fit is 0.47 within 25 miles of a Kalshi station but 0.15 beyond 100 miles, and only 7.4% of US PRF grids lie within 50 miles of one. Summer storm rain is too local for the tested airport gauges, so only 3 of the 36 two-month windows our farms insure can be copied well.
- **W5:** researched exotic payouts, such as exponential ones that pay little until a deep drought. For the exponential station payout, one-inch strikes reproduce 63% of variance; half-inch strikes would give 86%.

Beyond the plan (shared equally): new payouts and the advisor.

- **Payouts built to order.** Kalshi's listed monthly-rain ladder lets a portfolio pay a chosen amount on each listed strike bin. We designed insurance that pays little in a mild dry spell and a lot in a severe drought, instead of PRF's straight-line payout. Since it is built from listed contracts, a fully filled portfolio matches the payout on the station's rain bins. The farmer still bears station-to-farm risk.
- **Why drought-weighted.** The best contract for a risk-averse farmer under our stated wealth, risk-aversion and 20% price-load assumptions puts most payout into the driest months. Smooth drought-weighted shapes retain 90 to 93% of the modeled optimum's value; PRF's straight-line shape retains 67%. A $6,000 November contract at Dallas has a modeled fair premium of $1,205 and pays $3,727 if the airport gets at most 1 inch, $862 at 1 to 2 inches, and almost nothing above. The saved order book fills only 13% of that order near fair value.
- **Advisor.** A farmer types one sentence ("150 acres of hay near Weatherford, TX"), and Claude uses our tools to locate the land, check the nearest station, and design and explain a contract. The rule-based path runs in the demo; the Claude path has scripted-client tests but has not been tested against the live API.

2. Are the final results ready?

Almost, but not yet. We have enough for the presentation: the app designs and quotes payouts. A formal report needs more research on what these contracts are worth to real farms and how they compare with subsidized insurance. Our plan:

Owner	Research for the final report (due 12/15)
Jack	Model real-world farms accurately and backtest them: what each would pay for this insurance, and what it would have paid out in past years
Maya	Formalize the gap between our insurance and subsidized PRF: how much of it is the subsidy, how much is basis risk, and how much is liquidity
Evan	Clean up the app and research more exotic contract ideas, such as two-month and combined drought-and-flood contracts

All three of us will build the slides for 11/17 and write the final report.

3. Refinements to our plan

- **Scope:** instead of copying PRF, we are testing drought-weighted cover for months a farmer left unselected. Limited depth, basis risk, and price remain constraints.
- **Responsibilities:** updated to the roles in Section 2.